Modern Academy
for Engineering and Technology in Maadi
Basic Science Department

MODERN ACADEMY FOR ENGINEERING & TECHNOLOGY

Academic Year	2024-2025
Semester	First
Exam Date	12/11/2024
Academic Level:	Sophomore

MODEL ANSWER FOR THE MID-TERM WRITTEN EXAM

Subject: (MTHn103) **Mathematics-3** Differential Equations and Transforms
Spec.: Electrical and Mechanical Engineering
Examiners: Prof. Ashraf T.EL-Sayed and Dr. Amira M.Salem
Time: 60 minutes

Number of Pages: 1 **(Both Sides)** Number of Questions: 2 Attempt all questions

Question 1: [LO's: LO 1, LO 2, LO 4] **(12 points)**

Find the general solution of the first ordinary differential equations:

a) $e^{-y}\frac{dy}{dx}=e^{2-y}\left(\sec(y)+\frac{x^2}{\cos(y)}\right)$. **(3 points)**

$$e^{-y}\frac{dy}{dx}=\frac{e^2}{e^y}\left(\sec(y)+\sec(y)x^2\right)$$

$$e^{-y}\frac{dy}{dx}=\frac{e^2}{e^y}\sec(y)\left(1+x^2\right)$$

$$\frac{e^{-y}e^{y}}{\sec(y)}dy=e^2\left(1+x^2\right)dx$$

$$\int\cos(y)dy=\int e^2\left(1+x^2\right)dx$$

$$\boxed{\sin(y)=e^2\left(x+\frac{x^3}{3}\right)+C}$$

b) $x\cos\left(\frac{y}{x}\right)y'=y\cos\left(\frac{y}{x}\right)-x\sin\left(\frac{y}{x}\right)$. **(3 points)**

$$\frac{y}{x}=u\rightarrow y=xu\rightarrow y'=xu'+u$$

$$\cos(u)(xu'+u)=u\cos(u)-\sin(u)$$

$$x\cos(u)u'+\cancel{u\cos(u)}=\cancel{u\cos(u)}-\sin(u)$$

$$x\cos(u)\frac{du}{dx}=-\sin(u)$$

$$\int\frac{\cos(u)}{\sin(u)}du=\int-\frac{1}{x}dx$$

$$\ln|\sin(u)|=-\ln|x|+\ln C\rightarrow\ln|\sin(u)|=\ln\left|\frac{C}{x}\right|$$

$$\boxed{\sin(u)=\frac{C}{x}\rightarrow\sin\left(\frac{y}{x}\right)=\frac{C}{x}}$$

c) $\left(2xy-9x^2\right)dx+\left(2y+x^2+1\right)dy=0$. **(3 points)**

$M_y=2x \qquad N_x=2x$

$M_y=N_x \ \left(Exact\right)$

$\int\left(2xy-9x^2\right)dx+\int\left(2y+x^2+1\right)dy=C$

$x^2y-3x^3+y^2+\cancel{x^2y}+y=C$

$\boxed{x^2y-3x^3+y^2+y=C}$

d) $xy'+4y=x^4y^2$. **(3 points)**

$y'+\frac{4}{x}y=x^3y^2$

$P(x)=\frac{4}{x} \qquad Q(x)=x^3 \qquad n=2$

$P^*(x)=\frac{-4}{x} \quad Q^*(x)=-x^3 \quad 1-n=-1$

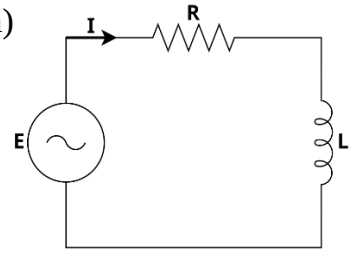
$\mu=e^{\int\frac{-4}{x}dx}=e^{-4\ln x}=e^{\ln x^{-4}}=x^{-4}$

$y^{-1}=x^4\left[\int -x^{-1}dx+C\right]$

$\boxed{y^{-1}=x^4\left[-\ln|x|+C\right]}$

Question 2: [LO's: LO 1, LO 2, LO 4] **(8 points)**

a)



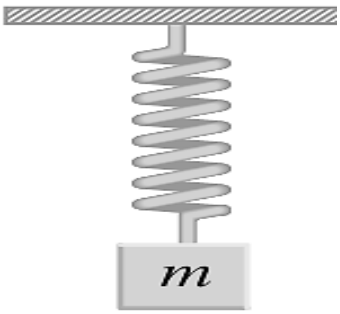
$0.02\frac{dI}{dt}+0.06I=0.02e^{4t}$ **(4 points)**

$\frac{dI}{dt}+3I=e^{4t}$

$\mu=e^{\int 3dt}=e^{3t}$

$I=e^{-3t}\left[\int e^{7t}dt+C\right]$

$\boxed{I=e^{-3t}\left[\frac{e^{7t}}{7}+C\right]}$

b) 

$$2\frac{d^2x}{dt^2}+128x=0$$

(4 points)

$$\frac{d^2x}{dt^2}+64x=0$$

$$m^2+64=0$$

$$m=\pm 8i$$

$$\boxed{x=c_1\cos(8t)+c_2\sin(8t)}$$

GOOD LUCK